

XINGE GUO

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RESEARCH INTERESTS

Computer vision and machine learning for medical image analysis, with a focus on **Representation Learning**, **Image Segmentation**, **Clinical Decision Making** and **Treatment Planning**.

EDUCATION

Duke University, Durham, NC, USA

Aug 2025 – Present

M.S. in Biomedical Engineering · Advisor: Prof. Sina Farsiu (VIP Lab)

GPA: 3.81 / 4.00

- **Fellowships:** Research Fellowship for BME MS/MEng Students (Fall 2025 & Spring 2026); Merit-based Scholarship (Fall 2025)

Shenzhen University, Shenzhen, China

Sep 2021 – Jun 2025

B.Eng. in Biomedical Engineering, Class of Excellence

GPA: 3.91 / 4.50

- **Honors:** Top Innovative Talent Scholarship (Dec 2022, Dec 2023, Dec 2024); Full Scholarship for Class of Excellence (Sep 2021)

PUBLICATIONS

1. He, C.; Zhang, D.; Xiao, F.; **Guo, X.**; Feng, J.; Wang, Y.; Farsiu, S. Toward Multimodal Understanding in Low-Level Vision: The Rise of Vision-Language Models. Preprints 2026, 2026061854. Under review at **IEEE Trans. Pattern Analysis and Machine Intelligence (TPAMI)** · Preprints, 2026
2. Xiao F, Hu P, Xu L, **Guo X**, Qin G, Shen Y, Fang C, Zhang R, He C, Farsiu S. "Beyond Ground-Truth: Leveraging Image Quality Priors for Real-World Image Restoration." **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.**
3. Dai J, **Guo X**, Zhang H, et al. "Cone-beam CT Landmark Detection for Measuring Basal Bone Width: A Retrospective Validation Study." **BMC Oral Health, 2024**, 24(1): 1091. (Co-First Author)

RESEARCH EXPERIENCE

Xu Lab, Computational Biology Department, Carnegie Mellon University

Jun 2026 – Present

Research Intern · Advisor: Prof. Min Xu

Pittsburgh, PA, USA

Project 1 — Vessel Segmentation on Digital Subtraction Angiography (Ongoing)

- Developed deep learning models for blood vessel segmentation on digital subtraction angiography (DSA) images to support downstream vascular analysis.

Project 2 — Mechanistic Interpretability of Medical Vision Transformers (Ongoing)

- Used mechanistic interpretability and sparse feature clustering to reveal that medical vision transformers rely substantially on non-pathological but causally dominant (NPC) features, explaining their poor localization of pathology (manuscript in preparation).

Vision and Image Processing (VIP) Laboratory, Duke University

Sep 2025 – Present

Graduate Researcher · Advisor: Prof. Sina Farsiu

Durham, NC, USA

Project 1 — Leveraging Image Quality Priors for Real-World Image Restoration (CVPR 2026)

- Co-developed IQPIR, a framework that replaces pixel-level ground-truth supervision with a **learned image quality prior** to guide real-world image restoration, addressing the systematic bias of ground-truth-based metrics in optimizing restoration models.

Project 2 — Segmentation Difficulty Benchmark via VLM (Ongoing)

- Building a benchmark for **predicting segmentation difficulty from input images alone**.

- Designing a VLM-based agent that interacts with multiple segmentation models to estimate per-image difficulty and identify likely failure regions before deployment.

Project 3 — Hyperspherical Feature Representations for Segmentation (Ongoing)

- Exploring **hyperspherical feature representations** as a geometric prior on the embedding space to improve segmentation accuracy and robustness.

Department of Orthopaedic Surgery, Duke University School of Medicine

Dec 2025 – Present

Research Collaborator · Advisor: Dr. Brian C. Lau, MD

Durham, NC, USA

Project — Deep Learning-Based Glenoid and Humerus Bone Loss Measurement and Severity Stratification on 3D CT in Shoulder Instability (Ongoing)

- Developing 3D CT segmentation and quantitative measurement pipelines for glenoid and humeral bone loss assessment, in close collaboration with orthopaedic surgeons. Manuscript in preparation.

Medical Ultrasound Engineering Lab (National-Regional Key), Shenzhen University

Mar 2024 – Apr 2025

Undergraduate Researcher · Advisor: Prof. Chien Ting Chin

Shenzhen, China

Project — Left Ventricle Landmark Detection (Graduation thesis)

- Developed a lightweight, end-to-end deep learning network reformulating LV landmark detection.

Department of Radiology and Nuclear Medicine, Erasmus University Medical Center

Jul 2024 – Sep 2024

Research Intern · Mentor: Dr. Xianjing Liu

Rotterdam, the Netherlands

Project — GWAS on AI-Derived Phenotypes from Structural Brain MRI in UK Biobank

- Implemented a 3D convolutional autoencoder for representation learning on ~40,000 structural brain MRI scans from the UK Biobank, supporting downstream genome-wide association analyses on AI-derived imaging phenotypes.

Medical AI Laboratory, School of Biomedical Engineering, Shenzhen University

Aug 2022 – Oct 2024

Undergraduate Researcher · Advisor: Prof. Bingsheng Huang

Shenzhen, China

Project — CBCT Landmark Detection for Basal Bone Width Measurement (BMC Oral Health 2024, Co-First Author)

- Developed a U-Net-based deep learning model for precise orthodontic landmark detection on CBCT images.

SELECTED COURSEWORK

- **Duke (Graduate):** AI Image Assessment (Independent Study with Prof. Farsiu, A+), Signal Processing and Applied Math (A), Ultrasound Imaging (A), Digital Image Processing (Fall 2026, in progress)
- **Shenzhen U (Undergraduate):** Principles of Medical Imaging (A+), Medical Ultrasound Technology (A), Medical Signal Processing (A), Signals and Systems (A), Digital Electronics (A+), Probability and Statistics (A), Linear Algebra (A+), Calculus (A+), Python Programming (A)

TECHNICAL SKILLS

- **Programming:** Python (primary), MATLAB, C/C++, VHDL, Assembly
- **Languages:** English (Fluent, IELTS 7.5), Mandarin (Native), Cantonese (Native)

OTHER

- **Piano Performance:** Associate Diploma (ALCM), London College of Music, University of West London (Apr 2022).